

ARTURO ESPINOSA VARGAS

Cloud AI Engineer Google Cloud Python Vertex AI

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LinkedIn: [linkedin.com/in/arturo-espinosa-vargas-694a70239](https://www.linkedin.com/in/arturo-espinosa-vargas-694a70239) | GitHub: github.com/R2D2EV | Spanish: Native | English: Upper-Intermediate

PROFESSIONAL SUMMARY

Google Cloud Certified Associate Cloud Engineer and AI-focused software engineer with experience building Python applications, cloud data pipelines and evaluated generative AI systems. Built a multilingual RAG platform with Vertex AI, Cloud Run, Cloud SQL with pgvector and Cloud Storage. Combines software architecture and quality engineering with seven years of technical teaching and workshop delivery.

CORE SKILLS

Google Cloud: Vertex AI, Cloud Run, Cloud SQL for PostgreSQL, Cloud Storage, Cloud Build, Artifact Registry, Secret Manager, IAM and IAP; hands-on work with Compute Engine, GKE, VPC, load balancing, Cloud Logging and Cloud Monitoring.

Machine learning and AI: RAG, embeddings, Gemini, semantic retrieval, evaluation, grounding, citations and abstention; scikit-learn and Keras fundamentals.

Software and data: Python, FastAPI, REST APIs, SQL, PostgreSQL, pgvector, Pydantic, SQLAlchemy, ETL/ELT, JavaScript/Node.js and Power BI.

Delivery and quality: Docker, Kubernetes, Terraform, Git, Jenkins, CI/CD, pytest, Ruff, mypy, Robot Framework, Selenium WebDriver and structured logging.

PROFESSIONAL EXPERIENCE

Solera

QA Automation and AI Engineer | 2022 - August 2026

- Built automated suites with Robot Framework, Selenium WebDriver, Appium and Cypress, reaching approximately 85% test coverage across supported applications.
- Automated Jira and Tempo ETL pipelines to Excel and Power BI, reducing report preparation from days to hours.
- Improved Jenkins CI/CD execution efficiency by approximately 20% and developed QA dashboards for release planning and sprint health.
- Designed and deployed AI automation agents on AWS and Azure, including knowledge bases, model configuration and deployment.

SELECTED GOOGLE CLOUD AND AI PROJECTS

GroundedDocs RAG on Google Cloud

Python, Vertex AI, Cloud Run, Cloud SQL, Cloud Storage | 2026

- Designed and deployed a multilingual RAG application using FastAPI, Vertex AI embeddings and Gemini, Cloud Storage, and Cloud SQL with pgvector.
- Implemented document ingestion, semantic retrieval, source citations, relevance-based abstention, health checks and structured logging through a hexagonal architecture.
- Achieved Recall@5, MRR and F1 of 1.0 on a curated evaluation set and retained a 0.67 relevance threshold after retrieval experiments.
- Automated delivery from GitHub through Cloud Build quality gates and Artifact Registry to Cloud Run; managed runtime access with IAM, Secret Manager and IAP.

AI Daily Brief and Quiz Automation

n8n, Gemini, RSS, Gmail | 2026

- Built a scheduled n8n workflow that ingests AI news from RSS feeds, uses Gemini to generate ten concise learning briefs and five quiz questions, and delivers a structured digest through Gmail.

ADDITIONAL EXPERIENCE

UNAM, Faculty of Sciences | Teaching Assistant | 2019 - Present: Support Analytical Geometry courses for 60+ students, explain technical concepts and design reasoning-focused assignments.

Bootcamp Institute | Python Instructor | 2025: Delivered project-based training in automation, testing and debugging.

CERTIFICATIONS AND EDUCATION

Google Cloud Certified: Associate Cloud Engineer | Issued August 17, 2026; expires August 17, 2029

Additional credentials: Microsoft Certified: Azure AI Fundamentals (May 2026); Google Cloud Computing Foundations Certificate (March 2026)

Education: B.Sc. in Physics, National Autonomous University of Mexico (UNAM)

In progress: Google Cloud Professional Cloud Architect preparation through Google Get Certified 3; Google TRI AI Program